One-off masonry heaters

DIMENSIONING DOCUMENT

IN APPLICATION OF STANDARD EN 15544:2023

INPUT DATA

PROJECT DESCRIPTION	
Reference & Filename	PROJET EXEMPLE 15544
Date	01/01/2025
Country	France

TECHNICAL SPECIFICATIONS		
name	**symbol**	**value**
Nominal heat output	P_n	4.99 kW
storage period / nominal heating time	t_n	12.00 h
maximum load	m_B	18.46 kg
minimum load	m_B_min	9.23 kg
required minimum efficiency	n_min	78.00 %
Facing type		Without air gap
Core material		Firebrick with density 1750 to 2300 kg/m3, porosity 17 to 33%, thermal conductivity 0.90 to 1.35 W/m.K

FIREBOX SPECIFICATIONS		
Firebox type		Traditional
Type of appliance		Wood-burning heating appliance
Firebox depth (B)	B	44.0 cm
Firebox width (A)	A	42.0 cm
Firebox height (H)	H	78.0 cm
Ash pit height (AF)	AF	5.0 cm
Pressure loss coefficient from door (ζ)		0.300
Total air intake surface area on door		170.0 cm²
Glass width		15.0 cm
Glass height		20.0 cm

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REQUIREMENTS NOT MET

name	symbol	value	?	details
storage period / nominal heating time	t_n	12.00 h	OK	
maximum load	m_B	18.46 kg	OK	
minimum load	m_B_min	9.23 kg	OK	
glass area of the firebox	glass_area	300 cm²	OK	
firebox base	A_BR	□ 42.0 cm x 44.0 cm	OK	
firebox height	H_BR	78.00 cm	OK	
air-fuel ratio	λ	2.95	OK	
height of the lowest opening	height_of_the_lowest_opening	5.00 cm	OK	
		()	OK	

PRESSURE REQUIREMENTS (EN 15544)

somme des pertes de charges (Σ pr + Σ pu)	23.37 Pa	
somme des forces ascensionnelles (Σ ph)	23.95 Pa	
différence de pression (Σ ph - (Σ pr + Σ pu) >= 0.00 Pa)	0.58 Pa	OK
différence de pression (Σ ph - (Σ pr + Σ pu) <= 1.17 Pa)	0.58 Pa	OK

TEMPERATURE REQUIREMENTS (EN 15544)

température de la paroi du conduit de fumée à son sommet	85.1 °C	OK (>= 45.0 °C)

EMISSIONS AND EFFICIENCY

		regulation	?
désignation du foyer	standing standard burning chamber		
Efficiency	>= 82.1 %		
rendement minimum à puissance réduite (foyer couplé avec accumulateur)	>= 88.7 %		
rendement saisonnier	72.1 %	≥ 65.0 %	OK
Carbon monoxide (CO) à 13 % O2	1137 mg/Nm³	≤ 1500 mg/Nm³	OK
Dust à 13 % O2	26 mg/Nm³	≤ 40 mg/Nm³	OK
Organic gaseous compounds (OGC) à 13 % O2	87 mg/Nm³	≤ 120 mg/Nm³	OK
Nitrogen oxides (NOx) à 13 % O2	113 mg/Nm³	≤ 200 mg/Nm³	OK
Dust + OGC à 13 % O2	113 mg/Nm³	≤ 150 mg/Nm³	OK
organisme accredité ou notifié	Test Laboratory for Combustion Systems - Technical University of Vienna		
Local regulations	Label Flamme Verte	OK	
Country	France		
Type of appliance	Wood-burning heating appliance		

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FLUE GAS KEY VARIABLES	
température en sortie d'accumulateur	198.7 °C
tirage nécessaire	8.7 Pa
débit massique des fumées	64.6 g/s

EXIT TEMPERATURES ESTIMATION	
firebox mean temperature (in °C)	700.0 °C
mean temperature of the flue gas at the outlet of the firebox	550.0 °C
température en sortie d'accumulateur	198.7 °C
température à la sortie du conduit de fumée	151.1 °C
température de la paroi du conduit de fumée à son sommet	85.1 °C

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FLUE GAS PATH

type	id	name	length	geom	h	α	//	R th.	air space	loc.	t amb.	t	ρ	v	ζ	pu	kf	pRs	pRg	pH
Air intake	-	-	-	-	-	-	-	-	-	-	-	0.0 °C	-	-	-	4.00 Pa	-	-	-	-
Combustion air	0	porte	-	◯ 14.7 cm	-	-	1	-	-	Space where heater is located	-	0.0 °C	1.273 kg/m³	2.82 m/s	0.30	1.52 Pa	-	-	-	-
-> SOUS-TOTAL			0.000 m		0.000 m										0.30	1.52 Pa		-	-	-
Firebox	0	ascension dans foyer	0.780 m	▯ 42.0 cm x 44.0 cm	0.780 m	-	1	-	-	Space where heater is located	-	700.0 °C	0.354 kg/m³	0.99 m/s	-	-	2 mm	0.01 Pa	-	7.03 Pa
-> SOUS-TOTAL			0.780 m		0.780 m										0.00	-		0.01 Pa	-	7.03 Pa
[Chimney connector 1]																				
Chimney connector	0	sortie foyer	0.317 m	▭ 25.1 cm x 23.0 cm	-	-	1	-	-	Space where heater is located	-	537.2 °C	0.425 kg/m³	2.63 m/s	-	-	3 mm	0.08 Pa	-	-
Chimney connector	1	virage avant descente	-	▭ 25.1 cm x 23.0 cm	-	90.0 °	1	-	-	Space where heater is located	-	524.7 °C	0.432 kg/m³	2.59 m/s	1.20	1.83 Pa	-	-	-	-
Chimney connector	2	section geometry change	-	▭ 25.1 cm x 22.5 cm	-	-	1	-	-	Space where heater is located	-	524.7 °C	0.432 kg/m³	2.65 m/s	0.03	0.05 Pa	-	-	-	-
Chimney connector	3	descente	0.815 m	▭ 25.1 cm x 22.0 cm	−0.815 m	-	1	-	-	Space where heater is located	-	493.9 °C	0.449 kg/m³	2.60 m/s	-	-	3 mm	0.22 Pa	-	−6.58 Pa
Chimney connector	4	virage avant banc avant	-	▭ 25.1 cm x 22.0 cm	-	90.0 °	1	-	-	Space where heater is located	-	464.8 °C	0.467 kg/m³	2.51 m/s	1.20	1.77 Pa	-	-	-	-
Chimney connector	5	section geometry change	-	▭ 23.5 cm x 23.0 cm	-	-	1	-	-	Space where heater is located	-	464.8 °C	0.467 kg/m³	2.56 m/s	0.03	0.05 Pa	-	-	-	-
Chimney connector	6	banc avant	1.792 m	▯ 22.0 cm x 24.0 cm	-	-	1	-	-	Space where heater is located	-	406.9 °C	0.507 kg/m³	2.42 m/s	-	-	3 mm	0.48 Pa	-	-
Chimney connector	7	virage avant bout du banc	-	▯ 22.0 cm x 24.0 cm	-	90.0 °	1	-	-	Space where heater is located	-	356.2 °C	0.547 kg/m³	2.24 m/s	1.20	1.95 Pa	-	-	-	-
Chimney connector	8	section geometry change	-	▯ 21.0 cm x 24.0 cm	-	-	1	-	-	Space where heater is located	-	356.2 °C	0.547 kg/m³	2.34 m/s	0.07	0.11 Pa	-	-	-	-
Chimney connector	9	bout du banc	0.437 m	▯ 20.0 cm x 24.0 cm	-	-	1	-	-	Space where heater is located	-	344.8 °C	0.558 kg/m³	2.41 m/s	-	-	3 mm	0.14 Pa	-	-
Chimney connector	10	virage avant banc arrière	-	▯ 20.0 cm x 24.0 cm	-	90.0 °	1	-	-	Space where heater is located	-	333.8 °C	0.568 kg/m³	2.37 m/s	1.20	1.96 Pa	-	-	-	-
Chimney connector	11	section geometry change	-	▯ 19.5 cm x 24.0 cm	-	-	1	-	-	Space where heater is located	-	333.8 °C	0.568 kg/m³	2.43 m/s	0.04	0.06 Pa	-	-	-	-
Chimney connector	12	arrière banc	2.073 m	▯ 19.0 cm x 24.0 cm	-	-	1	-	-	Space where heater is located	-	286.1 °C	0.616 kg/m³	2.30 m/s	-	-	3 mm	0.68 Pa	-	-
Chimney connector	13	virage avant vers remontée	-	▯ 19.0 cm x 24.0 cm	-	90.0 °	1	-	-	Space where heater is located	-	245.3 °C	0.665 kg/m³	2.13 m/s	1.20	1.46 Pa	-	-	-	-
Chimney connector	14	section geometry change	-	▯ 20.0 cm x 24.0 cm	-	-	1	-	-	Space where heater is located	-	245.3 °C	0.665 kg/m³	2.03 m/s	0.05	0.06 Pa	-	-	-	-
Chimney connector	15	vers remontée	0.437 m	▯ 21.0 cm x 24.0 cm	-	-	1	-	-	Space where heater is located	-	237.4 °C	0.675 kg/m³	1.90 m/s	-	-	3 mm	0.10 Pa	-	-
Chimney connector	16	virage avant remontée	-	▯ 21.0 cm x 24.0 cm	-	90.0 °	1	-	-	Space where heater is located	-	229.9 °C	0.685 kg/m³	1.87 m/s	1.20	1.66 Pa	-	-	-	-

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Chimney connector	17	section geometry change	-	□ 21.0 cm x 23.0 cm	-	-	1	-	-	Space where heater is located	-	229.9 °C	0.685 kg/m³	1.95 m/s	0.06	0.09 Pa	-	-	-	-
Chimney connector	18	remontée	0.980 m	□ 21.0 cm x 22.0 cm	0.980 m	-	1	-	-	Space where heater is located	-	213.7 °C	0.708 kg/m³	1.98 m/s	-	-	3 mm	0.27 Pa	-	5.43 Pa
-> SOUS-TOTAL			6.851 m		0.165 m										7.48	11.04 Pa		1.96 Pa	-	−1.15 Pa
[Channels 2]																				
Channels	0	conduit simple peau 1	0.409 m	◯ 20.0 cm	0.409 m	-	1	0.001 m²K/W	-	Heated	0.0 °C	195.6 °C	0.735 kg/m³	2.80 m/s	-	-	1 mm	0.18 Pa	-	2.16 Pa
Channels	1	coude angle vif 30°	-	◯ 20.0 cm	-	30.0 °	1	-	-	-	-	193.0 °C	0.739 kg/m³	2.78 m/s	0.30	0.86 Pa	-	-	-	-
Channels	2	conduit simple peau 2	0.707 m	◯ 20.0 cm	0.612 m	-	1	0.001 m²K/W	-	Heated	0.0 °C	188.2 °C	0.747 kg/m³	2.75 m/s	-	-	1 mm	0.30 Pa	-	3.16 Pa
Channels	3	coude angle vif 30°	-	◯ 20.0 cm	-	30.0 °	1	-	-	-	-	184.2 °C	0.753 kg/m³	2.73 m/s	0.30	0.84 Pa	-	-	-	-
Channels	4	conduit simple peau 2	0.241 m	◯ 20.0 cm	0.241 m	-	1	0.001 m²K/W	-	Heated	0.0 °C	182.3 °C	0.756 kg/m³	2.72 m/s	-	-	1 mm	0.10 Pa	-	1.22 Pa
-> SOUS-TOTAL			1.357 m		1.262 m										0.60	1.70 Pa		0.58 Pa	-	6.54 Pa
[Chimney 3]																				
Chimney	0	intérieur	0.550 m	◯ 20.0 cm	0.550 m	-	1	0.051 m²K/W	-	Heated	0.0 °C	177.5 °C	0.765 kg/m³	2.69 m/s	-	-	1 mm	0.23 Pa	-	2.74 Pa
Chimney	1	combles	0.560 m	◯ 20.0 cm	0.560 m	-	1	0.051 m²K/W	-	Exterior	0.0 °C	170.5 °C	0.777 kg/m³	2.65 m/s	-	-	1 mm	0.23 Pa	-	2.73 Pa
Chimney	2	extérieur	0.900 m	◯ 20.0 cm	0.900 m	-	1	0.051 m²K/W	-	Exterior	0.0 °C	161.1 °C	0.793 kg/m³	2.59 m/s	-	-	1 mm	0.36 Pa	-	4.24 Pa
Chimney	3	extérieur (ajout)	0.400 m	◯ 20.0 cm	0.400 m	-	1	0.051 m²K/W	-	Exterior	0.0 °C	153.4 °C	0.808 kg/m³	2.55 m/s	-	-	1 mm	0.16 Pa	-	1.83 Pa
Chimney	4	element terminal	-	◯ 20.0 cm	-	-	1	-	-	-	-	151.1 °C	0.812 kg/m³	2.53 m/s	0.60	1.56 Pa	-	-	-	-
-> SOUS-TOTAL			2.410 m		2.410 m										0.60	1.56 Pa		0.99 Pa	-	11.54 Pa
=> TOTAL			11.398 m		4.617 m										8.98	19.83 Pa		3.54 Pa	0.00 Pa	23.95 Pa

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SOLID FUEL BURNING APPLIANCES (DONNÉES UTILES POUR EN 13384-1)		
Name		Traditional + EN 15544
Type of appliance		Wood-burning heating appliance
rendement de l'appareil (à puissance utile nominale)	ηWN	82.1 %
rendement de l'appareil (à puissance utile réduite)	ηWmin	88.7 %
concentration de CO2 à puissance nominale (% en volume sur fumées sèches)	σ_CO2	7.1 %
concentration de CO2 à puissance réduite (% en volume sur fumées sèches)	σ_CO2	7.1 %
teneur en vapeur d'eau des fumées à puissance nominale (% en volume)	σ_H2O	8.4 %
teneur en vapeur d'eau des fumées à puissance réduite (% en volume)	σ_H2O	8.4 %
Nominal heat output	QN	49.19 kW
Reduced heat output	Q min	24.59 kW
Flue gas temperature at nominal burn rate	TWN	198.7 °C
Flue gas temperature at reduced burn rate	TW min	132.5 °C
débit massique des fumées (nominale)	ṁ	64.6 g/s
débit massique des fumées (réduite)	ṁ min	32.3 g/s
débit massique d'air comburant (nominale)	ṁB	61.1 g/s
débit massique d'air comburant (réduite)	ṁB min	30.6 g/s
débit volumique des fumées (nominale)		318.54 m³/h
débit volumique des fumées (réduite)		136.91 m³/h
débit volumique de l'air comburant (nominale)		172.87 m³/h
débit volumique de l'air comburant (réduite)		86.43 m³/h
fonctionnement en pression		sous pression négative
tirage minimal requis	PW	8.66 Pa
tirage maximal requis	PWmax	9.83 Pa

GEOGRAPHICAL LOCATION	
Altitude	128 m
Within 12 miles of an ocean or Great Lake	Non
CHIMNEY LOCATION ON ROOF	
Chimney height above ridgeline	> 16 in.
ADJACENT BUILDINGS	
Horizontal distance between chimney and adjacent buildings	> 50 ft.

PENALTY		
Penalty	P_L	0 Pa

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TEMPERATURE REQUIREMENTS (EN 13384-1)			
Burn rate		NOMINAL	
température de condensation des fumées	T_sp	41.5 °C	
température limite	T_ig	41.5 °C	
température à la sortie du conduit	T_ob	165.4 °C	
température de la paroi à la sortie du conduit	T_iob	93.3 °C	NA
No condensation (T_iob - Tig >= 0)	T_iob - Tig	51.8 °C	NA

PRESSURE REQUIREMENTS (EN 13384-1)		
fonctionnement en pression	sous pression négative	
Burn rate	NOMINAL	
————————		
résultante de pression à l'alimentation en air (P_B) - Minimum draft	4.00 Pa	
résultante de pression à l'alimentation en air (P_B) - Maximum draft	4.00 Pa	
tirage minimal (P_Z)	8.99 Pa	
tirage minimal requis (P_Ze)	8.40 Pa	
tirage maximal (P_Zmax)	9.36 Pa	
tirage maximal admis (P_Zemax)	9.52 Pa	
————————		
P_Z - P_Ze >= 0	0.58 Pa	NA
P_Z - P_B >= 0	4.99 Pa	NA
P_Zemax - P_Zmax >= 0	0.16 Pa	NA

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OUTSIDE TEMPERATURE REFERENCE		
symbol	**conditions**	**value**
Exterior air temperature (T_L)	Minimum draft	0.0 °C
Exterior air temperature (T_L)	Maximum draft	0.0 °C
Flue gas temperature at chimney exit (T_uo)	Minimum draft	T_L
Flue gas temperature at chimney exit (T_uo)	Maximum draft - Dead air space - Wet (with water vapor)	−15.0 °C
Flue gas temperature at chimney exit (T_uo)	Maximum draft - Dead air space - Dry (without water vapor)	0.0 °C
Flue gas temperature at chimney exit (T_uo)	Maximum draft - Ventilated air space - Wet (with water vapor) - hauteur zones non-chauffées <= 5m	0.0 °C
Flue gas temperature at chimney exit (T_uo)	Maximum draft - Ventilated air space - Wet (with water vapor) - hauteur zones non-chauffées > 5m	−15.0 °C
Flue gas temperature at chimney exit (T_uo)	Maximum draft - Ventilated air space - Dry (without water vapor) - hauteur zones non-chauffées <= 5m	0.0 °C
Flue gas temperature at chimney exit (T_uo)	Maximum draft - Ventilated air space - Dry (without water vapor) - hauteur zones non-chauffées > 5m	0.0 °C

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